

Unit 8 – Inference for Categorical Data: Chi-Square
Notes 4 – Chi-Square Test for Association/Independence

Chi-Square Test for Association/Independence

- This test is very similar to the chi-square test for homogeneity, but the difference is important:
 - _____ the data comes from impacts the conclusions we can draw

In a chi-square test for homogeneity, the data is gathered by taking _____ random samples, and therefore the conclusions we draw relate to how those data are distributed across the different populations (or treatments).

In a chi-square test for association/independence, the data is gathered by taking _____ random sample, and therefore the conclusions we draw relate to how the two variables are related.

Chi-Square Test for Homogeneity	Chi-Square Test for Association/Independence
H_0 : There is no difference in the distribution of a categorical variable for several populations or treatments.	H_0 : Two categorical variables are _____ in the population of interest. OR H_0 : There is _____ between the two categorical variables in the population of interest
H_A : There is a difference in the distribution of a categorical variable for several populations or treatments.	H_A : Two categorical variables are _____ in the population of interest OR H_A : There is _____ between two categorical variables in the population of interest

- The conditions for this test are the same for the chi-square test for homogeneity.
- The expected counts, chi-square statistic, degrees of freedom, and p-value are all calculated the same way as the chi-square test for homogeneity.
- The conclusions and follow-up analysis will be interpreted in the same manner, but adjusted for the new hypothesis.

The Three Chi-Square Tests

GOF Test	Homogeneity	Association/Independence
Is the claimed distribution for a single categorical variable what is observed?	Is the distribution of a categorical variable the same across several populations?	Is the distribution of two categorical variables associated in the same population of interest?
Data comes from one random sample with multiple categories	Data comes from two or more random samples	Data comes from one random sample with the individuals classified according to two categorical variables

Example

What kinds of people do not use seat belts? A random sample of 459 people asked if they wear a seat belt while driving and how many cigarettes they smoke per day. The study wanted to test the claim that the amount of smoking is independent of seat belt use. This is a theory based on the idea that people who smoke more are less concerned about their health and safety and are therefore less inclined to wear seat belts. Is this theory supported by the data? Use a 0.05 significance level.

Observed Counts	Number of Cigarettes Smoked Per Day			
	0	1–14	15–34	35 and over
Wear Seat Belts	175	20	42	6
Doesn't Wear Seat Belts	140	18	43	15